Supplementary Material: PaTH-VLA: Progress-and-Target Context for Long-Horizon Robot Policy Adaptation

Anonymous CVPR submission

A. Pre-registration

The following text was frozen (SHA-256 `6e1d9523...`) before the first optimizer update and before the first scored episode. It is reproduced verbatim except for the removal of machine paths.

Frozen on 2026-11-08 (UTC) before any training run and before any scored evaluation episode of this study. Executor: Claude (Fable 5.1) on behalf of Qinzhen Ma; cluster hkust-cluster (8x H100 PCIe, GPUs 2/3 faulted and unused).

Question Does lightweight auxiliary supervision about *where the policy is in the task* (stage prediction) and *what it should attend to* (bounding-box prompts + target-location prediction) improve a pretrained generalist manipulation policy on long-horizon RoboCasa365 tasks, and which component carries the effect?

Models (official RoboCasa365 checkpoints, pinned in the weights index) • Xiaomi Robotics 1 (XR1)
• Awomo-0.5I
• Awomo-0.5V (Awomo-I/V are related variants: two model families, three models)

Tasks (10, fixed; selected 2026-11-05 from published per-task success in [20%, 60%] for all three models)
• composite-seen (training demos available): DeliverStraw, GetToastedBread, PackIdenticalLunches, SearingMeat, StirVegetables
• composite-unseen (never used for training or tuning): ArrangeTea, CuttingToolSelection, RecycleBottlesByType, WaffleReheat, WeighIngredients

Training conditions (per model, identical everything except the switches) | id | stage loss | box prompts in training input | location loss | |—-|————|————————————-|————| | S | on | off | off | | BL | off | on | on | | F | on | on | on | Shared: same official checkpoint init; frozen encoders/base weights; rank-8 LoRA on the same four late DiT projections plus stage head (16 classes) and location head (3 cameras x 2); identical demonstrations (20 train episodes per seen task = 100, SHA256-ranked, fixed on 2026-11-06), identical annotated window panel, identical sampling order (seed 20261106), microbatch 1 x accumulation 8, AdamW lr 1e-4 wd 0.01, identical optimizer update count within a model (set from a timing calibration *before* any run; the same count is then used for S, BL and F of that model). Loss = action flow-matching MSE (+ Awomo future-latent MSE) + 0.1 stage CE (if on) + 0.1 location smooth-L1 (if on). No stage text is added to the policy instruction in any condition. The box is a 2-px red rectangle drawn on the current frame only; history and future-prediction targets stay clean. No action-only retrained baseline (user decision).

Annotations Automatic pseudo-labels: Grounding DINO (base) proposals → SAM 2.1 (hiera-small) tight mask boxes → Qwen3-VL-4B-Instruct set-of-mark selection of the described target instance per camera (or "not visible"). No human review. Windows whose target is not visible in any camera (and the 'task complete' stage) are excluded from the panel for ALL arms. A random audit sample of the final labels is inspected visually and its hit rate reported.

Evaluation (primary) • Ordinary inputs for every condition: original task instruction, clean RGB, state. No feedback of predicted heads.
• Conditions evaluated per model: official checkpoint (reference), S, BL, F → 12 policies.
• 10 tasks x 20 fixed seeds (0-19) per policy = 200 episodes per policy, 2,400 total. Same seed ⇒ same initial scene
(checked by first-frame hash); official per-task horizon and official success function; policy query every 16 env steps (native chunking); policy RNG seed 4242 + replan index. • Primary endpoints (per model, paired by (task, seed), pooled over the 10 tasks):
(a) F - BL : contribution of stage supervision given box+location, (b) F - S : contribution of box+location given stage. Secondary: F - official (total short-budget adaptation, confounded with extra action training), S - official, BL - official; seen vs unseen subsets; per-task tables. • Statistics: paired success-rate difference with 95% CI from a stratified (by task) paired bootstrap (10,000 draws),
exact two-sided McNemar/sign test on discordant pairs, Holm correction over the two primary contrasts per model. A contrast counts as "supported" only if the Holm-corrected interval excludes 0 and the point estimate is $\geq$ +5 pp. • All episodes are reported (success / failure / error / timeout). No task, seed or checkpoint is replaced after seeing

results. Only the final checkpoint of each run is evaluated.

Optional secondary study (only if time permits, reported separately) Same F checkpoint with self-generated stage/box feedback vs ordinary inputs; not part of the primary claims.

Compute window Training and evaluation on hkust-cluster GPUs 0,1,4,5,6,7 (user authorized exclusive use on 2026-11-08). Deadline-driven: CVPR 2027 registration 2026-11-10 23:59 AoE, paper 2026-11-16 23:59 AoE.

B. Target definitions

Every stage description in the five training tasks (198 distinct strings) is mapped to a detector class, an instance hint for the selector, and a role by the rules below; three strings ("task complete") have no target and their windows are excluded for all arms. No string was left unresolved.

Stage kind	Pattern	Detector class	Instance hint / X as written / excluded
pick	"pick up the X from the Y"	X (pan→frying pan, bread→slice of bread)	the X about to
place	"place the X in/on the Y"	Y: tupperware→plastic food container, burner→stove burner, pan→frying pan, pot→cooking pot, plate, glass cup	the destination
execute	"open the drawer …"	drawer handle	"the front-left
execute	"turn on the P burner"	stove knob	interaction par
execute	"stir the vegetables in the pot"	cooking pot	the knob contr
press / wait	"push down the lever", "wait for the toaster …"	toaster lever	interaction par
navigate	"navigate to the X"	cup→glass cup, plate, fridge→refrigerator, tupperware→plastic food container, stove, counter→kitchen counter	interaction par X as written / r
done	"task complete"	—	excluded

Grounding DINO (base) runs with box and text thresholds 0.2, NMS 0.65 and at most six proposals per camera; SAM 2.1 (hiera-small) refines each proposal with a box prompt and the tight box of the mask replaces the proposal. Qwen3-VL-4B-Instruct sees the three cameras upscaled to 512×512 with numbered coloured boxes, the task instruction, the stage text and the instance hint, and returns one candidate number (or -1) per camera with greedy decoding. Windows: 8,931 training windows were queued, 8,174 have a confirmed target in at least one camera and form the shared panel (8,174 windows); 757 were excluded (606 of them navigation stages whose target was out of view).

C. Label audit

A random sample of 48 panel windows (seed 7; 8 per task kind, stratified by nothing else) was rendered with the selected box in each camera and inspected by the authors' assistant after labelling and before any training result was known. The 48 windows contain 103 camera views with a visible-target label.

- 93 of 103 boxes (90%) bound the described target. Eight of these bound the whole appliance rather than the named part (the toaster instead of its lever, the stove top instead of one burner); their centre is still on the correct object.
- 10 boxes are wrong: six bound a robot link or the gripper (typically when the arm occludes the target), three bound the wrong instance or object among duplicates (a container holding fish instead of the empty one; a plate instead of the burner), one bounds unrelated furniture.
- Views labelled "not visible" were not audited for misses; navigation stages have the highest miss rate because the target is often outside the field of view, which is why 606 of the 757 excluded windows are navigation windows.

These rates characterise the supervision the BL and F arms actually received; the location loss is therefore trained on roughly nine correct targets in ten. The review sheets (`ann/review/*.png`) and the per-window decisions are released with the code.

D. Training details and curves

Adapter placement. For XR-1 the adapters sit on the last four `attn.qkv_proj`/`attn.o_proj` projections of the action DiT, for Awomo-0.5I on the last four `to_q`/`to_v`/`to_qkv_mlp_proj` projections of its image-action DiT, for Awomo-0.5V on the last four `self_attn.q`/`self_attn.v` projections of its video-action DiT. Rank 8, A initialised Kaiming-uniform, B zero, so every arm starts as the exact official policy (verified by comparing zero-update exports with recorded original actions). The stage head is a $d\times16$ linear layer and the location head a $d\times6$ linear layer followed by a sigmoid, both on the hooked feature.

Window construction. A window at anchor t (multiple of 16, $t+32$ inside the episode) contains the history frames each model natively consumes (XR-1: $t-6, t-4, t-2, t$; Awomo-0.5I: every 40th frame back to $t-360$ plus the frame $t+16$ as the future image target; Awomo-0.5V: every 4th frame from $t-96$ to $t+32$, the future part being the video target), the corresponding states, the 32 future actions in the raw LeRobot layout, the official `subtask_idx` at t and, for BL and F, the pseudo-label box drawn on the three current frames only. Actions are converted to each model's native layout and normalisation exactly as in its official evaluation code; Awomo-V targets are resampled from 20 Hz to 15 Hz with nearest-neighbour gripper and mode channels, as in its release.

Optimisation. AdamW, learning rate 10^{-4}, weight decay 0.01, gradient clipping as configured by LeRobot's default preset, one window per micro-batch, eight micro-batches per update, 1,000 updates, data order fixed by the seed

20261106 and identical across arms. The flow-matching time $t \sim \mathcal{U}(0, 1)$ and the noise are drawn from a seed that depends only on the micro-batch index, so S, BL and F see identical noise. Wall-clock per run on one H100 shared with evaluation workers: 41 minutes for XR-1, 107–173 minutes for the Awomo models. Figure 1 shows the loss terms.

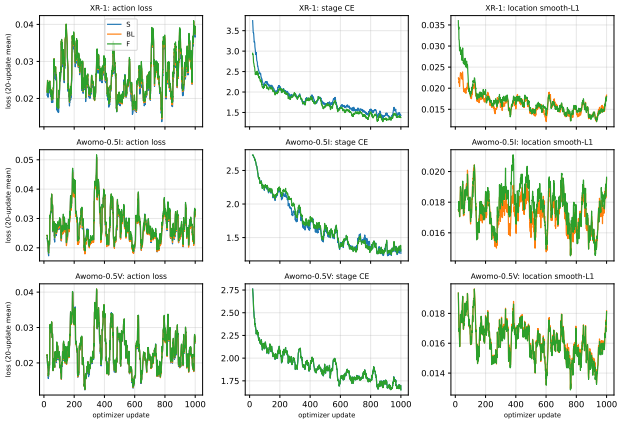


Figure 1. Training-loss terms per model and arm (20-update running mean). Stage cross-entropy and location smooth-L1 are only shown for the arms in which they are switched on.

E. Per-task results and episode-level data

Table **??** of the main paper lists every per-task success count. Table 1 splits each paired contrast into the five composite-seen tasks, whose demonstrations were used for adaptation, and the five composite-unseen tasks. Table 2 gives episode statistics and Table 3 the rescue/regression counts of the full arm against the official checkpoint.

Episode-level data. Each of the 2,400 episodes is logged as one JSON record with model, condition, task, seed, instruction, horizon, success flag, step count, number of policy queries, median query latency, the SHA-256 of the first observation (used to verify that all four policies of a model saw the same initial scene), the first and last agent-view frames, and a 5 fps agent-view video. The analysis script recomputes every number in the paper from these records.

References

Model	Contrast	seen (100 pairs) Δ (pp)	seen (100 pairs) 95% CI	unseen (100 pairs) Δ (pp)	unseen (100 pairs) 95% CI
XR-1	F−BL	+4	[-7, +16]	-2	[-11, +7]
	F−S	+4	[-6, +14]	-2	[-10, +7]
	F−official	+13	[+2, +24]	+2	[-6, +11]
	S−official	+9	[-2, +20]	+4	[-4, +12]
	BL−official	+9	[-2, +20]	+4	[-3, +12]
Awomo-0.5I	F−BL	+2	[-8, +13]	+5	[-4, +15]
	F−S	-1	[-13, +11]	-2	[-12, +8]
	F−official	+0	[-11, +10]	-13	[-24, -2]
	S−official	+1	[-11, +13]	-11	[-22, +0]
	BL−official	-2	[-15, +11]	-18	[-29, -7]
Awomo-0.5V	F−BL	-4	[-15, +7]	-3	[-13, +7]
	F−S	-12	[-21, -3]	-5	[-15, +4]
	F−official	-14	[-25, -3]	-8	[-20, +4]
	S−official	-2	[-14, +10]	-3	[-13, +7]
	BL−official	-10	[-21, +1]	-5	[-16, +5]

Table 1. Paired contrasts split by task group (composite-seen tasks whose demonstrations were used for adaptation vs. composite-unseen tasks). Task-stratified paired bootstrap, 4,000 draws.

Model	Policy	median steps (success)	timeouts	mean wall-clock (s)
XR-1	official	1384	132	200
	S	1427	119	226
	BL	1332	119	206
	F	1281	117	228
Awomo-0.5I	official	1314	111	326
	S	1351	121	322
	BL	1585	131	353
	F	1397	124	328
Awomo-0.5V	official	1376	112	385
	S	1422	117	380
	BL	1319	127	394
	F	1271	134	376

Table 2. Episode statistics. Every failure is a horizon timeout (no early terminations occurred). Wall-clock includes simulation and policy inference on shared GPUs.

Model	F rescues	F regressions	net
XR-1	33	18	+15
Awomo-0.5I	24	37	-13
Awomo-0.5V	27	49	-22

Table 3. Episodes the F arm solves that the official checkpoint fails (rescues) and vice versa (regressions), out of 200 paired episodes.